Conditional probability

1 Find $P(B|A)$ given the following probabilities for an experiment in which A and B are two possible events:

$$P(A \cap B) = 0.48 \text{ and } P(A) = 0.6$$

2 A basketball team has a probability of 0.8 of winning its first season and 0.15 of winning its first season and its second season. Find the probability of winning the second season, given they won first.

3 In a population, 35% of people have brown eyes, 45% of people have blonde hair, and 10% of people have both brown eyes and blonde hair. A person is chosen randomly from the population. Find the probability that they have:

a Blonde hair, given that they have brown eyes.

b Brown eyes, given that they have blonde hair.

4 Find $P(A \cap B)$ given the following probabilities for an experiment in which A and B are two possible events:

$$P(B|A) = 0.8 \text{ and } P(A) = 0.5$$

5 The probability of two independent events, A and B are $P(A) = 0.6$ and $P(B) = 0.7$. Determine the probability of:

a Both A and B occurring

b Neither A nor B

c A or B or both

d B but not A

e A given that B occurs.

6 Amelia has 4 cards, each with one of the following numbers printed on it:

$$2, 9, 1, 7$$

a How many combinations of 3 cards can she take from them?

b If she arranges a particular group of 3 cards in a line, how many possible arrangements are there?

c Given that the arrangement contains 1, find the probability it starts with 2.

d Given that the arrangement contains 7, find the probability it starts with 9 or 1.

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7 A student is choosing 2 units to study at university: a language and a science unit. They have 5 languages and 8 science units to choose from.

a If they choose one of each, find the total number of combinations of choices.

b If Italian is one of the languages they can choose from, find the probability they choose Italian as their language.

c French is one of the available languages. Find the probability they choose French as their language given that they choose Chemistry as their science unit.

8 For breakfast each morning, Marge eats porridge, toast or cereal. With that she will either drink orange juice, tea or hot chocolate. Find the probability that Marge will:

a Eat toast and drink tea.

b Eat cereal or drink orange juice.

c Eat porridge given that she drinks hot chocolate.

9 A netball coach is choosing players for the Goal Keeper and Goal Defence positions out of the following people:

- **Goal Keeper position**: Beth, Amy, Joy, Tara.
- **Goal Defence position**: Eve, Cara, Daisy, Kim, Liz.

The selection for each position is made independently.

a Find the probability the coach will choose Amy and Daisy.

b Find the probability the coach will choose Amy or Daisy.

c If the coach chooses Joy, find the probability she will choose Kim.

d Find the probability the coach will choose Eve given that Beth won't play with her.

10 Two dice are rolled, one after another. Find the probability, in simplest form, of rolling:

a A pair of fours, given the first die is a four.

b A pair with a sum of 7 or more given that the first die is a two.

11 A card is randomly drawn from a standard 52 pack of cards. Find the probability that it's a jack, queen, king or ace if:

a No additional information is known.

b We know it's a 10, jack or queen.

c We know it's a 9 or a queen.

d We know it's not 2, 3, 4 or 5.

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12 Four identical balls labelled 1, 2, 3 and 4 are in a bag. Two balls are randomly drawn from the bag in succession and without replacement. Find the probability, in simplest form, that:

a The first ball is labelled 4 and the second ball is labelled 2.

b The sum of the numbers on the two balls is 5.

c The second ball drawn is 1 given that the sum of the numbers on the two balls is 5.

13 Two cards are randomly drawn without replacement from a deck of cards numbered from 1 to 20. Find the probability that the second card is:

a An even number given that the first card is a 10.

b Less than 5 given that the first one is a 14.

c A number divisible by 5 given that the first card is a 15.

14 A flight departs from Melbourne to Sydney. The following probabilities were found with regard to departure times and weather:

- The probability that the flight departs on time, given the weather is fine in Melbourne is 0.9.
- The probability that the flight departs on time, given the weather is not fine in Melbourne is 0.7.
- The probability that the weather is fine on any particular day in July is 0.4.

Find the probability that:

a The flight from Melbourne to Sydney departs on time in a day in July.

b The weather is fine in Melbourne given that the flight departs on time on a day in July.

15 A museum has new software to determine whether a painting is fake or authentic. The system has an accuracy of 75%. Art experts have previously determined that 5% of paintings that the museum examines are fakes.

Find the probability that:

a The software correctly identifies a painting as fake.

b The software incorrectly identifies an authentic painting as fake.

c The software will identify a painting as fake.

d A painting that is identified as fake is actually fake, correct to two decimal places.

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16 There are two positions available at a company and the applicants have been shortlisted down to 6: Patricia, Nadia, Amy, Aaron, Lachlan, Jimmy. The two people to fill the position will be picked randomly.

- **Event A:** Patricia is chosen
- **Event B:** Aaron is chosen

a Describe the probability $P(A|B)$.

b Is $P(A|B)$ less than, greater than or equal to $P(A \cap B)$?

17 Two sets of numbers, A and B, are such that set A contains the even numbers from 1 to 20, inclusive, and set B contains the factors of 48 from 1 to 20, inclusive.

a List the numbers in set A.

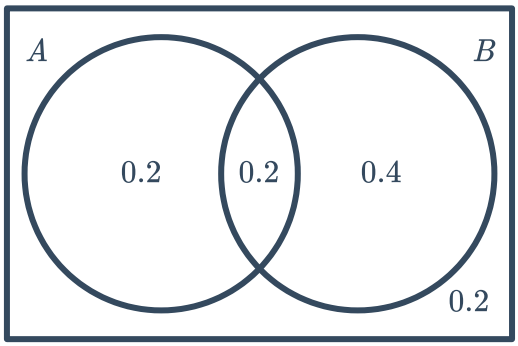
b List the numbers in set B.

c Find the total number of unique numbers across both sets.

d Find the probability that a randomly selected number is in set B, given that it is in set A.

Venn diagrams

18 Find $P(A|B)$ given the following probability Venn Diagram:



19 Consider the following probability Venn Diagram. Find:

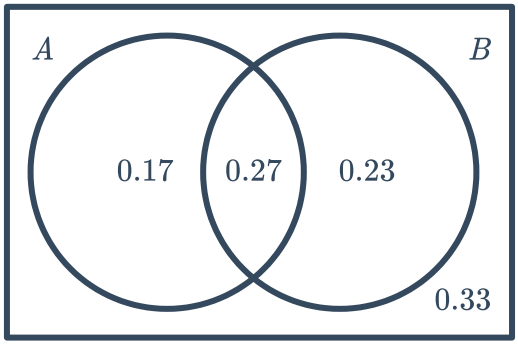
a $P(A' \cap B)$

b $P(A' \cup B')$

c $P(A|B)$

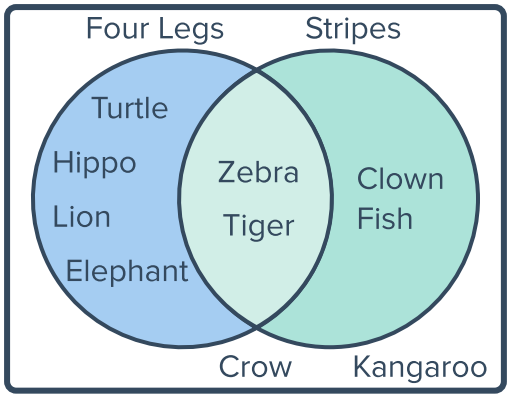
d $P(B|A')$

e $P(A \cup B|B')$



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20 A student creates the following diagram of their favourite animals:

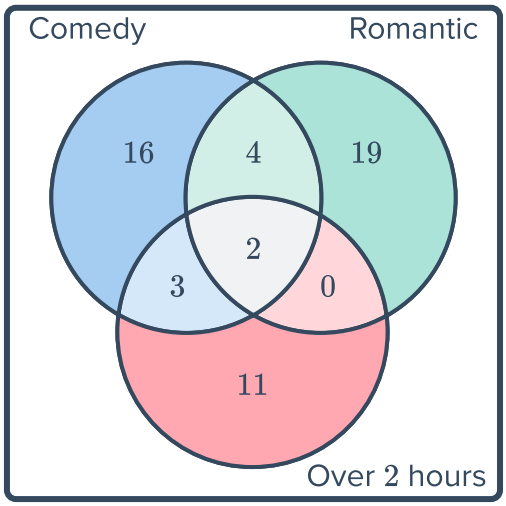


a How many of the animals have four legs?

b How many of the animals have four legs and stripes?

c How many of the animals have four legs or stripes, but not both?

21 Harry is struggling to decide what movie to watch. He decides to pick one at random from his collection. A Venn diagram of his collection sorts movies into three categories:



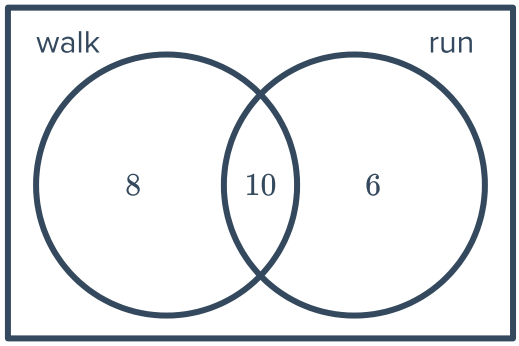
a How many movies are there in Harry's collection?

b Harry wants to watch a comedy. Find the probability he will select a a comedy.

c Harry is after a film that goes for longer than 2 hours. Find the probability that he will select a suitable film given that he chooses a comedy.

d Harry is after a film that goes for less than 2 hours. Find the probability that he will select a suitable film given that he selects a romantic comedy.

22 A group of people were asked what type of exercise they like to do. The results are shown in the Venn diagram below:
A person is chosen from the group at ramdom.



a Find the probability that the person walks.

b Find the probability that the person runs, given that they also walk.

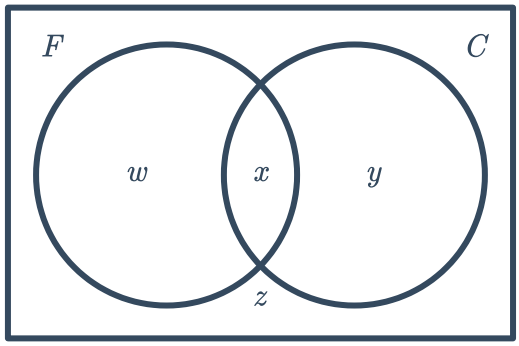
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23 At a university there are 816 students studying first year engineering, 497 of whom are female (set F). Of the 348 students studying Civil Engineering (set C), 237 of them are women.

a Find the value of:

i w **ii** x **iii** y **iv** z

b Find the probability that a randomly selected male student does not study Civil Engineering.



Tree diagrams and tables

24 Neil watches three episodes of TV each night. He begins with News or Current Affairs, then the next two shows are either Comedy, Horror or Animation. He never watches more than one Horror show, and if he watched the News, he will follow this immediately with a Comedy.

a Construct a tree diagram of all possible options.

b If, at every stage, the possible outcomes of each choice are equally likely, find the probability Neil watches two Comedies, given that he watched the News.

25 When Fred gets ready for work in summer, he first decides whether it will be a hot day (H) or not (N) and then decides whether to wear a tie (T) or to just dress casually (C).

The chance of Fred deciding it will be a hot day is 0.7. If he decides it will be a hot day, there is a 0.2 chance he will wear a tie. If he decides it will not be a hot day, there is a 0.85 chance he will wear a tie.

a Construct a probability tree for this situation.

b Calculate the probability Fred decided it was a hot day given that he wears a tie. Round your answer to two decimal places.

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26 A player is rolling 2 dice and looking at their difference, that is, the largest number minus the smaller number. They draw up a table of all the possible dice rolls for 2 dice and what their difference is:

a State the sample space for the difference of two dice.

b Find the probability the dice will have a difference of 2.

c Find the probability the dice will have a difference of 2 given that one of the dice rolled is a 3

	1	2	3	4	5	6
1	0	1	2	3	4	5
2	1	0	1	2	3	4
3	2	1	0	1	2	3
4	3	2	1	0	1	2
5	4	3	2	1	0	1
6	5	4	3	2	1	0

27 In a survey, 151 children were asked whether they had read or watched at least one of the stories in the Harry Potter series.

- 60 hadn't read any of the books
- 33 had read and watched at least one of the stories
- 85 had watched the movie,
- 8 had never read or watched any of these

a Complete the following table for these results:

	Read the book	Didn't read the book	Total
Watched the movie	33		85
Didn't watch the movie		8	
Total		60	151

b Find the probability that a randomly selected student had:

i Read the book but hadn't watched the movie.

ii Read the book given that they had watched the movie.

iii Seen the movie given that they hadn't read the book.

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28 Two hundred people were questioned about whether they voted for Labor, Liberal or Greens last election and who they'll vote for this election.

	Labor (next election)	Liberal (next election)	Greens (next election)	Total
Labor (last election)	26	40		70
Liberal (last election)			1	90
Greens (last election)	10		5	
Total		140	10	200

a Complete the table

b Find the probability that a randomly selected person will vote Labor next election given that they voted Liberal last election.

c Find the probability that a randomly selected person did not vote Green last election given that they will vote Green next election.

d If a person voted Labor or Liberal last election, find the probability they'll vote Liberal next election.

29 A class of 13 students had their height, shoe size and 400 m race time collected into the following table:

Athlete	400 m time (sec)	Height (cm)	Shoe size
Aaron	94	137	8
Tobias	62	165	10
Lisa	89	100	11
Georgia	64	155	7
Ben	98	100	10
Patricia	127	102	11
Vanessa	84	147	7
Sandy	98	144	8
Adam	54	154	7
Amelia	84	147	8
Rosey	81	126	12
Neil	104	134	9
Roald	67	127	8

a Let T be the event that a randomly chosen student from the class has a race time greater than 90. Find the value of $P(T)$.

b Let H be the event that a randomly chosen student from the class has a height greater than 130. Find the value of $P(H)$.

c Find the value of $P(T|H)$.

d Let S be the event that a randomly chosen student from the class has a shoe size greater than or equal to a size 10. Find the value of $P(H|S)$.

e Hence, which of the following events appear to be dependent, H and S or H and T?

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